Use Case Narratives: Parent Module

ASPIRE: After-School Program Integrated Resource & Education System — Parent Module

Module: Parent Module | Prepared by: Farhana Ahmed (2312079) | Course: CSE451

1. Overview

This document describes the use cases of the Parent module as step-by-step narratives. Each narrative is based on the Parent user stories (US-P1 to US-P7) in the SRS and uses the same terms. The parent has read-only access: no use case lets the parent edit academic, attendance or financial records.

Actors. The primary actor is the Parent. The Email Service is a secondary actor used for notifications. Teacher, Student and Center Admin do not interact with this module directly, but their actions create the data that the parent sees.

Common rule. Every use case except UC-P1 requires a logged-in parent, and the system checks on the backend that the requested child is linked to that parent.

2. Use Case Summary

ID	Use case	User story	Relationships
UC-P1	Log In and View Dashboard	US-P1	«extend» Switch Between Children.
UC-P2	View Child Progress	US-P2	None.
UC-P3	View Child Attendance	US-P3	None.
UC-P4	Track Assignment Status	US-P4	«include» Receive Notifications (for overdue assignments).
UC-P5	View Fee Invoices	US-P5	«extend» Download Invoice (PDF).
UC-P6	View Payment Records	US-P6	«extend» Download Receipt.
UC-P7	Receive Notifications	US-P7	«extend» Set Notification Preferences. Used by UC-P4 for overdue assignments.

3. Use Case Narratives

UC-P1: Log In and View Dashboard

Use Case ID	UC-P1

Use Case Name	Log In and View Dashboard
Related User Story	US-P1 (and common stories US-C1, US-C2)
Primary Actor	Parent
Secondary Actor	None
Description	The parent logs in with email and password and sees a dashboard that summarizes each linked child.
Trigger	The parent opens the ASPIRE login page.
Preconditions	● Parent has an active account in the system. ● The parent knows the registered email and password.
Postconditions	● The parent is authenticated and a session is started. ● The dashboard shows data of the parent's linked children only.
Main Flow	1. Parent opens the login page. 2. System shows the email and password fields. 3. Parent enters email and password and submits. 4. System validates the credentials and checks that the account is active. 5. System identifies the role as Parent and redirects to the Parent dashboard. 6. System retrieves the children linked to the parent. 7. System displays, for each child, the attendance rate, pending assignments, latest grade, outstanding fees and recent notifications.
Alternative Flows	**A1. More than one child linked:** at step 7 the system shows a child selector. The parent selects another child and the dashboard refreshes (see Switch Between Children). **A2. Forgot password:** the parent selects “Forgot password” and enters the email. The system sends a time-limited, single-use reset link by email. **A3. No linked child:** the system shows a message asking the parent to contact the center. **A4. Logout:** the parent logs out from any page. The system ends the session and returns to the login page.
Exception Flows	**E1. Wrong email or password:** the system shows “Invalid email or password” without saying which one is wrong. **E2. Too many failed attempts:** after 5 wrong passwords within 15 minutes the account is locked for a period of time and the system shows a message. **E3. Deactivated account:** the system refuses login and shows a message to contact the center.

Business Rules and Notes	• Passwords are stored hashed (bcrypt); the system runs over HTTPS. • The parent sees only the children linked to the account; this is checked on the backend. • The dashboard loads within 3 seconds under normal conditions. • The login page works in mobile browsers.
Relationships	«extend» Switch Between Children.

UC-P2: View Child Progress

Use Case ID	UC-P2
Use Case Name	View Child Progress
Related User Story	US-P2
Primary Actor	Parent
Secondary Actor	None
Description	The parent views the child's published marks, grade breakdown, overall progress and the teacher's feedback.
Trigger	The parent selects “Progress” for a child.
Preconditions	• Parent has an active account and is logged in (see UC-P1). • At least one child is linked to the parent account by the Center Admin or Super Admin.
Postconditions	• The child's published results and feedback are displayed. • No data is changed (read-only).
Main Flow	1. Parent selects a child (if more than one) and opens “Progress”. 2. System checks that the child is linked to the parent. 3. System retrieves the published results, grade breakdown, feedback and progress data. 4. System displays results per subject and per assessment. 5. System displays the grade breakdown by category (Homework, Quizzes, Exams) and the overall progress bar. 6. System displays a summary: average marks and completed versus missed tasks. 7. Parent opens an assignment or result to read the teacher's feedback. 8. System displays the feedback linked to that assignment or result.

Alternative Flows	**A1. Grade not published:** the system shows “Unpublished” instead of the grade. **A2. No feedback yet:** the system shows a message that feedback is not available. **A3. Different subject:** the parent selects another subject and the system updates the view.
Exception Flows	**E1.** Child is not linked to the parent: the system denies access and shows a permission message. No data is returned. **E2.** Data cannot be loaded (server or network error): the system shows an understandable error message and lets the parent try again.
Business Rules and Notes	● Only grades and feedback published by the teacher are visible. ● Feedback of other students is never visible. ● The parent cannot edit any result. ● The page loads within 3 seconds.
Relationships	None.

UC-P3: View Child Attendance

Use Case ID	UC-P3
Use Case Name	View Child Attendance
Related User Story	US-P3
Primary Actor	Parent
Secondary Actor	None
Description	The parent views the child's attendance records and attendance percentage.
Trigger	The parent selects “Attendance” for a child.
Preconditions	● Parent has an active account and is logged in (see UC-P1). ● At least one child is linked to the parent account by the Center Admin or Super Admin.
Postconditions	● Attendance records and percentage are displayed. ● No data is changed (read-only).

Main Flow	1. Parent selects a child and opens “Attendance”. 2. System checks that the child is linked to the parent. 3. System retrieves the child's attendance records. 4. System displays records by date, course and status (Present, Absent, Late, Excused). 5. System displays the attendance percentage. 6. Parent filters by date range (and optionally by course). 7. System updates the records and the percentage for the selected filter.
Alternative Flows	**A1. Low attendance:** if the percentage is below the defined threshold (value to be confirmed by the team), the system highlights it visibly. **A2. No records:** if no record matches the filter, the system shows an empty-state message.
Exception Flows	**E1.** Child is not linked to the parent: the system denies access and shows a permission message. No data is returned. **E2.** Data cannot be loaded (server or network error): the system shows an understandable error message and lets the parent try again.
Business Rules and Notes	• Attendance is read-only for the parent. • The percentage is calculated from the attendance recorded by teachers. • Only the linked child's data is returned.
Relationships	None.

UC-P4: Track Assignment Status

Use Case ID	UC-P4
Use Case Name	Track Assignment Status
Related User Story	US-P4
Primary Actor	Parent
Secondary Actor	None (notification of overdue work uses UC-P7)
Description	The parent sees the child's assignments and their status so that homework is completed on time.
Trigger	The parent selects “Assignments” for a child.
Preconditions	• Parent has an active account and is logged in (see UC-P1). • At least one child is linked to the parent account by the Center Admin or Super Admin.

Postconditions	• The child's published assignments and statuses are displayed. • No data is changed (read-only).
Main Flow	1. Parent selects a child and opens “Assignments”. 2. System checks that the child is linked to the parent. 3. System retrieves the published assignments of the child's enrolled courses (drafts are excluded). 4. System displays for each assignment the course, title, due date, marks and status (Pending, Submitted, Overdue). 5. System displays overdue assignments in a separate, highlighted area. 6. Parent filters the list by status or course. 7. System updates the list.
Alternative Flows	**A1. Assignment becomes overdue:** the system sends a notification to the parent (see UC-P7). **A2. No match:** if no assignment matches the filter, the system shows an empty-state message.
Exception Flows	**E1.** Child is not linked to the parent: the system denies access and shows a permission message. No data is returned. **E2.** Data cannot be loaded (server or network error): the system shows an understandable error message and lets the parent try again.
Business Rules and Notes	• Draft assignments are never visible. • The parent can view but cannot submit or edit anything for the child. • Statuses use UTC timestamps and are shown in Bangladesh time.
Relationships	«include» Receive Notifications (for overdue assignments).

UC-P5: View Fee Invoices

Use Case ID	UC-P5
Use Case Name	View Fee Invoices
Related User Story	US-P5
Primary Actor	Parent
Secondary Actor	None
Description	The parent views the child's fee invoices, amounts and due dates, and can download an invoice.
Trigger	The parent selects “Invoices” for a child.

Preconditions	• Parent has an active account and is logged in (see UC-P1). • At least one child is linked to the parent account by the Center Admin or Super Admin. • The Center Admin has generated at least one invoice (otherwise see A2).
Postconditions	• The child's invoices are displayed. • If requested, a PDF copy of the invoice is downloaded.
Main Flow	1. Parent selects a child and opens “Invoices”. 2. System checks that the child is linked to the parent. 3. System retrieves the invoices generated by the Center Admin. 4. System displays each invoice with invoice ID, course, billing period, total fee, discount, amount payable, due date and status (Paid, Partially Paid, Due, Overdue). 5. System highlights overdue invoices. 6. Parent opens an invoice to see its details.
Alternative Flows	**A1. Download invoice:** the parent selects “Download PDF”. The system generates a PDF that contains only that child's invoice and downloads it (see Download Invoice). **A2. No invoices:** the system shows a message that no invoice has been generated yet.
Exception Flows	**E1.** Child is not linked to the parent: the system denies access and shows a permission message. No data is returned. **E2.** PDF generation or download fails: the system shows an error message and lets the parent try again.
Business Rules and Notes	• The parent sees only invoices of own linked children. • Invoices are created and changed only by the Center Admin; the original fee information is kept for audit. • Payment is not made inside the system (recorded by the Center Admin).
Relationships	«extend» Download Invoice (PDF).

UC-P6: View Payment Records

Use Case ID	UC-P6
Use Case Name	View Payment Records
Related User Story	US-P6
Primary Actor	Parent
Secondary Actor	None

Description	The parent views the child's payment history, remaining balance and receipts.
Trigger	The parent selects “Payments” for a child.
Preconditions	● Parent has an active account and is logged in (see UC-P1). ● At least one child is linked to the parent account by the Center Admin or Super Admin.
Postconditions	● The child's payment records are displayed. ● If requested, a receipt is downloaded.
Main Flow	1. Parent selects a child and opens “Payments”. 2. System checks that the child is linked to the parent. 3. System retrieves the payments recorded by the Center Admin. 4. System displays each payment with date, amount, method, transaction ID and remaining balance. 5. System shows partial payments together with the updated outstanding balance. 6. Parent opens a payment to see its details.
Alternative Flows	**A1. Download receipt:** the parent selects “Download receipt”. The system generates the receipt and downloads it (see Download Receipt). **A2. No payments:** the system shows a message that no payment has been recorded yet.
Exception Flows	**E1.** Child is not linked to the parent: the system denies access and shows a permission message. No data is returned. **E2.** Receipt generation or download fails: the system shows an error message and lets the parent try again.
Business Rules and Notes	● The parent cannot edit or delete payment records. ● Payments are recorded only by the Center Admin and are never permanently deleted. ● Only payments of own linked children are visible.
Relationships	«extend» Download Receipt.

UC-P7: Receive Notifications

Use Case ID	UC-P7
Use Case Name	Receive Notifications
Related User Story	US-P7
Primary Actor	Parent

Secondary Actor	Email Service
Description	The parent is notified about important events of the child and can view, filter and manage notifications.
Trigger	An event occurs for a linked child: absence, overdue assignment, newly published grade or feedback, new invoice, upcoming due date, payment recorded, or a teacher or center announcement.
Preconditions	● Parent has an active account and is logged in (see UC-P1). ● At least one child is linked to the parent account by the Center Admin or Super Admin. ● The notification type is enabled in the parent's preferences (otherwise see A4).
Postconditions	● A notification is stored and shown in the parent's in-app list. ● An email is sent if the email channel is enabled. ● The delivery status (generated, sent, delivered, failed, read) is recorded.
Main Flow	1. System detects an event related to a linked child. 2. System creates a notification for the parent of that child. 3. System checks the parent's preferences for this notification type. 4. System delivers the in-app notification and, if enabled, sends an email through the Email Service. 5. Parent opens the notification list; the system shows notifications, newest first. 6. Parent opens a notification. 7. System marks it as read and shows the details with a link to the related record.
Alternative Flows	**A1. Filter:** the parent filters the list by notification type. **A2. Mark as read:** the parent marks one or more items as read from the list. **A3. Set preferences:** the parent chooses which notification types to receive (in-app or email) and the system saves the choice (see Set Notification Preferences). **A4. Type turned off:** the system does not deliver the notification.
Exception Flows	**E1. Email delivery fails:** the system records the status as failed and retries. The in-app notification stays available. **E2.** A notification failure never changes or deletes the original record (grade, invoice, assignment, announcement).
Business Rules and Notes	● Notifications are delivered within a few minutes of the event. ● A notification goes only to the parents of the relevant child and never exposes another family's information. ● Delivery status is tracked where the service supports it.
Relationships	«extend» Set Notification Preferences. Used by UC-P4 for overdue assignments.